WTW 158 Worksheet 7 on Differentiation

In order to be sure you have mastered differentiation you need to complete the following:

A. The 200$^+$ Challenge. This challenge consists of problems in the Study Guide under Technique Mastering, Pages 44 - 48, sections I to K. The solutions are available on ClickUP. Complete this at home.

B. TEXTBOOK PROBLEMS
p204 no 47, 59, 61, 63
p215 no 25, 27, 39, 49, 51, 75
p223 no 23, 31, 33, 37, 39, 45, 47
p264 no 1, 3, 33, 35, 51

C. If you have done all of the above and you need more challenging problems then you can try your hand at The Ultimate Differentiation Challenge, available on ClickUP under the tab Worksheets. This is not for everyone and you should only attempt it if you have truly mastered differentiation.

D. TEST QUESTIONS

Question 1

The equation of the tangent line to the graph of $y = e^{2x}$ at the point where $x = 0$ is

[a] $y = (2e^{2x})x + 1$	[b] $y = (2e^{2x})x$	[c] $y = 2x$	
[d] $y = x$	[e] $y = 2x + 1$	[f] $y = x + 1$	[g] None of these

Question 2

If $y^3 \sin x + \cos y = x$ then $\dfrac{dy}{dx} =$

[a] $\dfrac{1 - y^3 \cos x}{3y^2 \sin x - \sin y}$	[b] $\dfrac{1}{3y^2 \cos x + \cos y}$	[c] $\dfrac{1 - y^3 \cos x}{3y^2 \sin x}$
[d] None of these		

Question 3

Indien $f(x) = \cosh(e^{2x})$, dan is $f'(x) =$

If $f(x) = \cosh(e^{2x})$, then $f'(x) =$

[a] $\sinh(e^{2x})$	[b] $e^{2x}\sinh(e^{2x})$	[c] $-2e^{2x}\sinh(e^{2x})$	[d] $2e^{2x}\sinh(e^{2x})$
[e] Geen van hierdie/None of these			

E. SELECTED ANSWERS

A. Answers on the 200+ Challenge is available on ClickUP under Technique Mastering
B. Answers in textbook.
D. 1 [e], 2 [a] 3 [d]

F. Start on Unit 4.1 p283 no 27, 33, 39, 49, 55, 61

Worksheet 7 Solutions

A. The 200⁺ Challenge. See ClickUP (Solutions)

B. Textbook problems

p204 : 47, 59, 61, 63

47. $y' = -\sin(\sin 3\theta) \cdot \cos 3\theta \cdot 3$

$= -3\cos 3\theta \cdot \sin(\sin 3\theta)$

$y'' = (-3)(-\sin 3\theta)(3) \times \sin(\sin 3\theta) + (-3\cos 3\theta)[\cos(\sin 3\theta) \times \cos 3\theta \times 3]$

$= 9\sin 3\theta \sin(\sin 3\theta) - 9\cos^2 3\theta \cdot \cos(\sin 3\theta)$

59. $f(x) = 2\sin x + \sin^2 x$

$f'(x) = 2\cos x + 2\sin x\cos x$

Tangent is horizontal if $f'(x) = 0 \Rightarrow 2\cos x(1+\sin x) = 0$

$\Rightarrow \cos x = 0$ or $\sin x = -1$

$\Rightarrow x = \pi/2 + 2k\pi$ or $x = 3\pi/2 + 2k\pi$, $k \in \mathbb{Z}$

Then $f(\pi/2) = 3$

$f(3\pi/2) = -1$

Points on curve:

$(\pi/2 + 2k\pi, 3)$ and $(3\pi/2 + 2k\pi, -1)$

61. $F(x) = f(g(x))$

$F'(x) = f'(g(x)) \cdot g'(x)$

$F'(5) = f'(g(5)) \cdot g'(5)$

$= f'(-2) \cdot 6 = 4 \cdot 6 = 24$

62. $h(x) = \sqrt{4 + 3f(x)}$

$h'(x) = \frac{1}{2}(4+3f(x))^{-1/2}(3f'(x))$

$h'(\ ) = \frac{1}{2}(4+3f(1))^{-1/2}(3f'(1))$

$= \frac{1}{2}(4+3 \cdot 7)^{-1/2}(3)(4)$

$= 6/5$

p215 : 25, 27, 39, 49, 51, 75

25. $y'\sin 2x + y\cos 2x \cdot 2 = \cos 2y + (x)(-\sin 2y)(2)(y')$

$\Rightarrow y'[\sin 2x + 2x\sin 2y] = \cos 2y - 2y\cos 2x$

$\Rightarrow y' = \dfrac{\cos 2y - 2y\cos 2x}{\sin 2x + 2x\sin 2y}$

if $x = \pi/2$, $y = \pi/4$, then

$y' = \dfrac{0 - 2(\pi/4)(-1)}{0 + 2(\pi/2)(1)} = \dfrac{1}{2}$

tangent line:

$y - \pi/4 = \frac{1}{2}(x - \pi/2)$

$\Rightarrow y = \frac{1}{2}x$

27. $2x - [y + xy'] - 2yy' = 0$

$\Rightarrow y'[-x - 2y] = -2x + y$

$\Rightarrow y' = \dfrac{-2x+y}{-x-2y}$

If $x = 2$, $y = 1$ then

$y' = \dfrac{-2(2)+1}{-2-2(1)} = \dfrac{-3}{-4} = \dfrac{3}{4}$

tangent line:

$y - 1 = 3/4(x - 2)$

$y = 3/4x - \frac{1}{2}$

39. $y + xy' + e^y y' = 0$ — ①

So $y'(x + e^y) = -y$

$\Rightarrow y' = \dfrac{-y}{x+e^y}$ — ②

Diff ① again:

$y' + (y' + xy'') + e^y y' y' + e^y y'' = 0$

When $x = 0$ in $xy + e^y = e$ then $e^y = e$

Then $y = 1$ and $y' = \dfrac{-1}{e}$

in ②:

$$-\frac{1}{e}+\left(-\frac{1}{e}\right)+e\left(-\frac{1}{e}\right)\left(-\frac{1}{e}\right)+ey''=0$$

$$\Rightarrow ey''=\frac{1}{e} \Rightarrow y''=\frac{1}{e^2}$$

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49. $y=\tan^{-1}\sqrt{x}$

$$y'=\frac{1}{1+(\sqrt{x})^2}\times\frac{1}{2\sqrt{x}}$$

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51. $y=\sin^{-1}(2x+1)$

$$y'=\frac{1}{\sqrt{1-(2x+1)^2}}\times 2$$

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75. $x^2y^2+xy=2$

$$\Rightarrow x^2\cdot 2yy'+y^2\cdot 2x+xy'+y=0$$

$$\Rightarrow y'(2x^2y+x)=-2xy^2-y$$

$$\Rightarrow y'=\frac{-2xy^2-y}{2x^2y+x}$$

So $y'=\dfrac{-2xy^2-y}{2x^2y+x}=-1$

$$\Rightarrow 2xy^2+y=2x^2y+x$$

$$\Rightarrow y(2xy+1)=x(2xy+1)$$

$$\Rightarrow y(2xy+1)-x(2xy+1)=0$$

$$\Rightarrow (2xy+1)(y-x)=0$$

$\Rightarrow xy=-\frac{1}{2}$ or $y=x$

But: $xy=-\frac{1}{2}\Rightarrow$

$x^2y^2+xy=\frac{1}{4}-\frac{1}{2}\neq 2$ ×

So $x=y$.

Then $x^2y^2+xy=2$

$$\Rightarrow x^4+x^2=2$$

$$\Rightarrow x^4+x^2-2=0$$

$$\Rightarrow (x^2+2)(x^2-1)=0$$

$\Rightarrow x^2=-2$ impossible

$\therefore x^2=1 \Rightarrow x=\pm 1$

Points: $(-1,1)$ $(1,1)$

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p 223: 23, 31, 33, 37, 39, 45, 47

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23. $y=\sqrt{x}\ln x$

$$y'=\frac{1}{2\sqrt{x}}\ln x+\frac{\sqrt{x}}{x}$$

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31. $f(x)=\ln(x+\ln x)$

$$f'(x)=\frac{1+\frac{1}{x}}{x+\ln x}$$

$$f'(1)=\frac{2}{1+0}=2$$

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33. $y=\ln(x^2-3x+1)$ (3,

$$y'=\frac{2x-3}{x^2-3x+1}$$

$$y'(3)=\frac{3}{1}=3$$

tangent line:

$$y-0=3(x-3)$$

$$\Rightarrow y=3x-9$$

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37. $f(x)=cx+\ln(\cos x)$

$$f'(x)=c+\frac{-\sin x}{\cos x}$$

$$f'(\pi/4)=c-\tan(\pi/4)=$$

$$\Rightarrow c-1=6$$

$$\Rightarrow c=7$$

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39. $y=(2x+1)^5(x^4-3)^6$

$$\ln y=5\ln(2x+1)+6(x^4-3)$$

$$\frac{1}{y}y'=\frac{(5)(2)}{2x+1}+\frac{(6)(4x^3)}{x^4-3}$$

$$y'=y\left[\frac{10}{2x+1}+\frac{24x^3}{x^4-3}\right]$$

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45. $y=x^{\sin x}$

$$\ln y=\sin x\ln x$$

$$\frac{1}{y}y'=\cos x\ln x+\frac{\sin x}{x}$$

$$y'=y\left[\cos x\ln x+\frac{\sin x}{x}\right]$$

47. $(\cos x)^x = y$

$\Rightarrow \ln y = x \ln \cos x$

$\frac{1}{y} y' = \ln\cos x + x \cdot \frac{-\sin x}{\cos x}$

$y' = \ln \cos x - x\tan x$

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p 264 : 1, 3, 33, 35, 51

1(a) $\sinh 0 = \frac{1}{2}(e^0 - e^0) = 0$

(b) $\cosh 0 = \frac{1}{2}(e^0 + e^0) = 1$

3.(a) $\cosh(\ln 5)$

$= \frac{1}{2}(e^{\ln 5} + e^{-\ln 5})$

$= \frac{1}{2}(5 + \frac{1}{5}) = \frac{26}{10} = \frac{13}{5}$

(b) $\cosh 5 = \frac{1}{2}(e^5 + e^{-5})$

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33. $h(x) = \sinh(x^2)$

$h'(x) = \cosh(x^2) \times 2x$

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35. $g(x) = \cosh(\ln x)$

$g'(x) = \sinh(\ln x) \times \frac{1}{x}$

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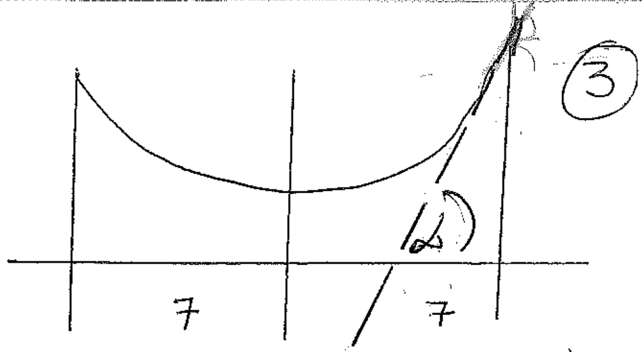
51. $y = 20\cosh\left(\frac{x}{20}\right) - 15$

a) $y' = 20 \sinh\left(\frac{x}{20}\right) \times \frac{1}{20}$

Right pole is positioned at $x = 7$. There fere

$y'(7) = \sinh\left(\frac{7}{20}\right) \simeq 0.3572$

b) If α is the angle between the tangent line and the x-axis then $\tan\alpha = \sinh\left(\frac{7}{20}\right)$



$\Rightarrow \alpha = \tan^{-1}\left(\sinh\frac{7}{20}\right)$

$\simeq 0.343$ rad

$\simeq 19.66°$

angle between line and pole is $90° - 19.66°$

$= 70.34°$

C. Ultimate Diff Challeng

D. Test Questions

Q1 [e]

$y = e^{2x}$: $x = 0,\ y = 1$

$y' = e^{2x} \cdot 2$ $\quad y'(0) = 2$

$\therefore\ y - 1 = 2(x - 0)$

$\Rightarrow y = 2x + 1$

Q2 [a]

$3y^2 \cdot y' \sin x + y^3 \cos x - (\sin y) y' = 1$

$y'[3y^2\sin x - \sin y] = 1 - y^3\cos x$

$y' = \frac{1 - y^3\cos x}{3y^2\sin x - \sin y}$

Q3 [d]

$f(x) = \cosh(e^{2x})$

$f'(x) = \sinh(e^{2x}) \times e^{2x} \times 2$

Unit 4.1

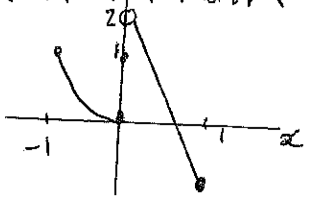
p 283 no 27, 33, 39, 49, 55, 61

27. $f(x) = \begin{cases} x^2 & \text{if } -1 \le x \le 0 \\ 2-3x & \text{if } 0 < x \le 1 \end{cases}$

No absolute / local max

absolute min. $f(1) = -1$

Local minimum $f(0) = 0$



33. $g(t) = t^4 + t^3 + t^2 + 1$

$g'(t) = 4t^3 + 3t^2 + 2t$

$= t(4t^2 + 3t + 2)$

$4t^2 + 3t + 2$ has no real solutions

$\therefore g'(t) = 0 \Rightarrow t = 0$

Only critical number is 0

39. $F(x) = x^{4/5}(x-4)^2$

$F'(x) = x^{4/5} \cdot 2(x-4) + (x-4)^2 \cdot \frac{4}{5}x^{-\frac{1}{5}}$

$= \frac{1}{5}x^{-\frac{1}{5}}(x-4)[5.x.2 + (x-4).4]$

$= \dfrac{(x-4)(14x-16)}{5x^{1/5}}$

$= \dfrac{2(x-4)(7x-8)}{5x^{1/5}}$

Critical numbers:

$F'(x) = 0 \Rightarrow x = 4, x = \frac{8}{7}$

$F'(0)$ does not exist $\therefore x = 0$

(4)

49. $f(x) = 2x^3 - 3x^2 - 12x + 1 \quad [-2,3$

$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2$

$= 6(x+1)(x-2) = 0$

$\Leftrightarrow x = -1, x = 2$

$\therefore f(-2) = -3, \quad f(-1) = 8$

$f(2) = -19, \quad f(3) = -8$

$f(-1) = 8$ absolute max

$f(2) = -19$ absolute min

55. $f(t) = t - \sqrt[3]{t} \quad [-1, 4]$

$f'(t) = 1 - \frac{1}{3}t^{-2/3} = 1 - \frac{1}{3t^{2/3}}$

$f'(t) = 0 \Leftrightarrow 1 = \frac{1}{3t^{2/3}} \Rightarrow t^{\frac{2}{3}}$

$t = \pm\left(\frac{1}{3}\right)^{3/2} = \pm\frac{1}{\sqrt{27}} = \pm\frac{1}{3\sqrt{3}} =$

$f'(t)$ does not exist if $t = 0$

$f(-1) = 0 \quad f(0) = 0$

$f\left(\frac{-1}{3\sqrt{3}}\right) = \frac{2\sqrt{3}}{9} \simeq 0.3849$

$f\left(\frac{1}{3\sqrt{3}}\right) = -\frac{2\sqrt{3}}{9}$

$f(4) = 4 - \sqrt[3]{4} \simeq 2.413$

So $f(4) = 4 - \sqrt[3]{4}$ is absolute max

$f\left(\frac{\sqrt{3}}{9}\right) = -\frac{2\sqrt{3}}{9}$ absolute mi

61. $f(x) = \ln(x^2 + x + 1) \quad [-1,$

$f'(x) = \dfrac{2x+1}{x^2+x+1} = 0 \Rightarrow x =$

Since

$x^2 + x + 1 > 0$ for $x \in \mathbb{R}$

$D_f = D_{f'}$

$f(-1) = \ln 1 = 0 \quad f(-\frac{1}{2}) = \ln$

$\simeq -0.$

$f(1) = \ln 3 \simeq 1.10$

$\therefore f(1) = \ln 3$ is absolute m

$f(-\frac{1}{2}) = \ln \frac{3}{4}$ is absolute min